

Statistics: Foundations and Methods

Foundations of Statistics

Statistics is the discipline of collecting, summarizing, and drawing conclusions from data in the presence of uncertainty. It is broadly divided into descriptive statistics, which summarize a dataset, and inferential statistics, which draw conclusions about a broader population from a sample.

Populations and Samples

A population includes every individual or item of interest, while a sample is a subset drawn from that population. Because it is rarely practical to measure an entire population, most statistical work relies on carefully drawn samples and quantifies the uncertainty introduced by sampling.

Descriptive Statistics

Measures of Central Tendency

The mean is the arithmetic average of a set of values. The median is the middle value when data is sorted, and is less sensitive to extreme outliers than the mean. The mode is the most frequently occurring value, and is especially useful for categorical data.

Measures of Spread

Variance measures the average squared deviation from the mean, and the standard deviation is its square root, expressed in the same units as the original data. The interquartile range, the difference between the 75th and 25th percentiles, describes spread in a way that is robust to outliers.

```
import numpy as np
data = [12, 15, 14, 10, 18, 22, 9]
print(np.mean(data), np.median(data), np.std(data))
```

Probability Fundamentals

Probability quantifies the likelihood of an event, expressed as a number between zero and one. The addition rule and multiplication rule govern how probabilities of combined events are calculated, and conditional probability describes how the likelihood of one event changes given knowledge of another.

Bayes' Theorem

Bayes' theorem relates the conditional probability of an event given evidence to the conditional probability of the evidence given the event, along with the prior probability of each. It underlies many modern approaches to updating beliefs as new data arrives.

Common Probability Distributions

- Normal distribution: symmetric, bell-shaped, describes many natural phenomena
- Binomial distribution: models the number of successes in a fixed number of independent trials
- Poisson distribution: models the number of events occurring in a fixed interval of time
- Exponential distribution: models the time between events in a Poisson process

Statistical Inference

Estimation

A point estimate is a single value used to approximate an unknown population parameter, such as using a sample mean to estimate a population mean. A confidence interval provides a range of plausible values for that parameter, along with a stated confidence level, typically ninety-five percent.

Hypothesis Testing

Hypothesis testing evaluates whether observed data provides enough evidence to reject a null hypothesis in favor of an alternative. The p-value represents the probability of observing data at least as extreme as what was measured, assuming the null hypothesis is true. A small p-value, conventionally below 0.05, is often treated as evidence against the null hypothesis, though this threshold is somewhat arbitrary and increasingly debated.

Type I and Type II Errors

A Type I error occurs when a true null hypothesis is incorrectly rejected, a false positive. A Type II error occurs when a false null hypothesis is not rejected, a false negative. There is generally a trade-off between the two: reducing one tends to increase the other for a fixed sample size.

Common Statistical Tests

t-tests

A t-test compares the means of one or two groups to determine whether any observed difference is likely to be due to chance. A paired t-test is used when measurements come from the same subjects under two conditions, while an independent t-test compares two separate groups.

Chi-square Test

The chi-square test evaluates whether there is a significant association between two categorical variables, by comparing observed frequencies in a contingency table to the frequencies expected if the variables were independent.

ANOVA

Analysis of variance, or ANOVA, extends the t-test to compare means across three or more groups simultaneously, testing whether at least one group mean differs significantly from the others.

```
from scipy import stats
group_a = [23, 25, 29, 22]
group_b = [31, 28, 30, 27]
stats.ttest_ind(group_a, group_b)
```

Correlation and Regression

Correlation measures the strength and direction of a linear relationship between two variables, ranging from negative one to positive one. It is important to remember that correlation does not imply causation; two variables can be correlated because of a shared underlying cause, a coincidence in a small sample, or genuine causal influence.

Linear Regression

Linear regression models the relationship between one or more independent variables and a continuous dependent variable by fitting a straight line, or hyperplane in higher dimensions, that minimizes the sum of squared errors between predicted and actual values.

```
import statsmodels.api as sm
X = sm.add_constant(x)
model = sm.OLS(y, X).fit()
print(model.summary())
```

Assumptions of Linear Regression

- Linearity between predictors and the outcome
- Independence of residuals
- Constant variance of residuals, known as homoscedasticity
- Approximately normally distributed residuals